

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	R KALAI AMUDHAN	Reg. No.:	192511292
Date:		Duration:	60 minutes

Code of Conduct

I R KALAI AMUDHAN (192511292) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R Kalai Amudhan

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

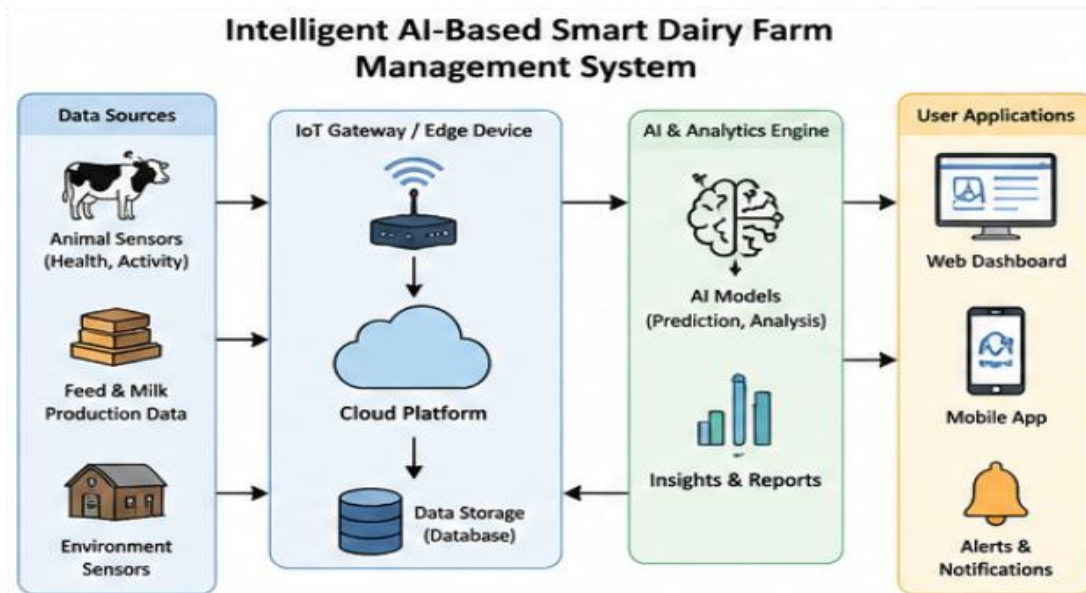
TEST CASE DESIGN

Problem Statement 17 – Intelligent AI-Based Smart Dairy Farm Management System

1. Introduction and System Understanding

The Intelligent AI-Based Smart Dairy Farm Management System is intended to support continuous livestock-health monitoring, disease prediction, feeding optimization, milk-production tracking, farm reporting and real-time analytics. The proposed solution combines wearable sensors, IoT connectivity, AI processing and cloud-based information management.

Testing must establish that the system behaves correctly under normal operating conditions as well as under invalid, boundary, exceptional and failure conditions. Because the system supports animal welfare and farm productivity decisions, incorrect information can have operational consequences. Therefore, this test design emphasizes correctness, reliability, data integrity, security and usability.



1.1 Main System Functions

- Continuous monitoring of cattle health using wearable sensor information.
- AI-based prediction of possible livestock disease risk.
- Optimization and management of cattle feeding schedules.
- Recording and tracking of milk production.
- Generation of farm performance reports.
- Real-time analytics for farm management decisions.
- Alerts and information supporting animal welfare.
- Information supporting operational efficiency and cost reduction.

1.2 Overall Testing Objectives

- Verify that each major function meets its intended requirement.
- Verify that invalid and incomplete data are safely handled.
- Verify threshold and boundary behaviour.
- Verify that data remains associated with the correct cattle.
- Verify that temporary failures do not create corrupt or duplicate records.
- Verify that protected information is accessible only to authorized users.
- Provide traceability between requirements, scenarios and test cases.

2. Functional Requirements – Detailed Test Basis

ID	Functional Requirement	Detailed Verification Points
FR-01	Continuous cattle health monitoring	Sensor registration; cattle association; current readings; historical readings; invalid and missing readings; stale data.
FR-02	AI disease prediction	Valid prediction inputs; prediction availability; risk classification; boundary risk; insufficient data; prediction result association.
FR-03	Feeding schedule optimization	Valid schedule creation; time validation; quantity validation; schedule update; conflict handling.
FR-04	Milk production tracking	Valid quantity; date/time; cattle association; duplicate prevention; historical totals.
FR-05	Farm performance reports	Valid date range; report contents; empty periods; totals; export/readability.
FR-06	Animal welfare improvement	Health alerts; alert visibility; acknowledgement; historical alert traceability.
FR-07	Operational cost reduction	Availability of feeding/production information; efficiency indicators; consistent records.
FR-08	Dairy productivity analytics	Current dashboard; historical trends; refresh behaviour; report/dashboard consistency.

2.1 Functional Requirement Acceptance Points

- Every cattle record should have a unique identity within the test data.
- Sensor information must be linked to the intended cattle record.
- Health alerts should identify the relevant animal and condition.
- AI prediction output should correspond to the supplied test dataset.
- Feeding schedules should retain the correct time and quantity information.
- Milk production totals should be calculated only from valid records.
- Reports should use the selected reporting period.
- Analytics should not display invalid or stale information as current information.

3. Non-Functional Requirements and Quality Testing

ID	Quality Area	Test Objective	Example Evidence
NFR-01	Performance	Check response under normal and increased workload.	Timed dashboard/report execution.
NFR-02	Reliability	Check recovery after temporary failures.	Logs before and after interruption.
NFR-03	Usability	Check whether common farm tasks are understandable.	Task observations/screenshots.
NFR-04	Security	Check authentication and authorization.	Allowed/denied access evidence.
NFR-05	Availability	Check accessibility of core functions.	Availability observations/logs.
NFR-06	Data integrity	Check accuracy and duplicate prevention.	Input vs stored vs report comparison.
NFR-07	Scalability	Check behaviour with larger data volume.	Load-test results.
NFR-08	Maintainability	Check error messages/logging support diagnosis.	Error/log evidence.

3.1 Risk-Based Testing Priority

Risk	Impact	Priority	Testing Response
Missed disease alert	Potential delayed intervention	Critical	Threshold, prediction, stale-data and alert tests.
False disease alert	Unnecessary intervention	High	Boundary and decision-table tests.
Wrong feeding schedule	Waste or poor nutrition	High	Boundary, invalid and conflict tests.
Incorrect milk total	Wrong productivity decision	High	Duplicate, correction and report-total tests.
Unauthorized data access	Security/privacy impact	High	Authentication and role tests.
Network interruption	Temporary monitoring gap	Medium/High	Failure and recovery tests.
Large data volume	Slow/unavailable analytics	Medium	Performance and scalability tests.

3.2 Non-Functional Test Questions

- How does the dashboard behave when several users access it simultaneously?
- What happens when the sensor stops transmitting for a period?
- Can a user without permission access a protected report?
- Does the system clearly distinguish stale readings from current readings?
- Does a large historical dataset affect report generation or dashboard usability?
- Do error messages help the user correct invalid data?

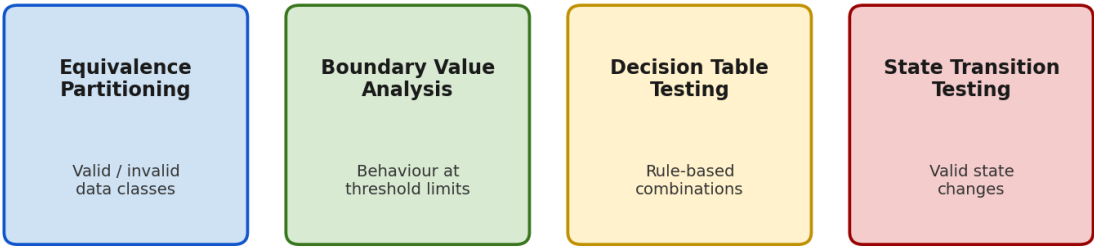
4. Test Scenarios – Expanded Scenario Catalogue

Scenario ID	Scenario	Verification	Requirement
TS-01	Cattle registration	Create a valid cattle profile and unique identifier.	FR-01
TS-02	Sensor association	Associate a registered sensor with the correct cattle.	FR-01
TS-03	Health reading	Receive and display valid sensor readings.	FR-01
TS-04	Invalid health data	Submit invalid or missing sensor data.	FR-01,NFR-06
TS-05	Stale health data	Allow a reading to become stale and verify status.	FR-01,NFR-02
TS-06	Disease prediction	Run AI prediction with complete valid data.	FR-02
TS-07	Prediction boundary	Evaluate risk at configured classification boundary.	FR-02
TS-08	Prediction missing data	Attempt prediction with insufficient input.	FR-02
TS-09	Health alert	Trigger a configured abnormal-health condition.	FR-02,FR-06
TS-10	Feeding schedule	Create a valid feeding schedule.	FR-03
TS-11	Feeding boundary	Test earliest/latest permitted schedule values.	FR-03
TS-12	Feeding invalid input	Reject invalid quantity/time.	FR-03
TS-13	Feeding conflict	Handle overlapping or conflicting schedules.	FR-03
TS-14	Milk entry	Record valid milk quantity.	FR-04
TS-15	Milk boundary	Test configured minimum/maximum quantity.	FR-04
TS-16	Milk invalid input	Reject negative, text or blank quantity.	FR-04
TS-17	Duplicate milk entry	Prevent unintended duplicate production record.	FR-04,NFR-06
TS-18	Farm report	Generate report for valid date range.	FR-05
TS-19	Invalid report range	Reject end date earlier than start date.	FR-05
TS-20	Empty report period	Generate report for a valid period without records.	FR-05
TS-21	Analytics refresh	Verify dashboard after new valid data.	FR-08
TS-22	Dashboard consistency	Compare dashboard totals with report totals.	FR-08,NFR-06
TS-23	Unauthorized access	Attempt protected function without permission.	NFR-04
TS-24	Authorized access	Access permitted function with valid account.	NFR-04
TS-25	Session expiry	Attempt protected operation after session expires.	NFR-04
TS-26	Recovery	Restart/reconnect after temporary failure.	NFR-02,NFR-05
TS-27	Concurrent users	Multiple users access dashboard simultaneously.	NFR-01
TS-28	Large dataset	Process high-volume cattle/sensor records.	NFR-07
TS-29	Report export	Export and compare report information.	FR-05
TS-30	Error handling	Trigger controlled invalid operation and inspect message.	NFR-03

5. Test Data Design and Equivalence Partitioning

The supplied assessment requires appropriate test data. The following data classes provide representative positive and negative partitions. Exact production limits should be taken from the final system specification during execution.

Test Design Techniques Applied



Field	Valid Partition	Invalid Partition	Boundary/Exceptional Partition
Cattle ID	Registered unique ID	Blank, duplicate, unsupported format	Maximum permitted length if specified
Sensor ID	Registered active sensor	Unknown, blank, duplicate association	Association change/reassignment
Temperature	Valid numeric current value	Text, blank, malformed	Configured health threshold
Heart rate	Valid numeric value	Text, blank	Configured alert boundary
Risk score	Valid AI score	Missing/invalid	Exact alert classification boundary
Milk quantity	Valid non-negative value	Negative/text/blank	Configured min/max
Feeding time	Valid time format	Invalid format	Earliest/latest permitted time
Report date	Valid start/end relationship	End before start	Same-day boundary
Credentials	Valid authorized credentials	Wrong/expired credentials	Session expiration boundary
Connectivity	Stable connection	No connection	Disconnect/reconnect transition

5.1 Equivalence Partitioning Test Cases

Case	Partition	Input	Expected Result
EP-01	Valid sensor data	Numeric current reading	Accepted.
EP-02	Invalid sensor data	Text reading	Rejected/flagged.
EP-03	Missing sensor data	Blank required field	Handled as unavailable/validation.
EP-04	Valid milk	18 L	Accepted.
EP-05	Negative milk	-2 L	Rejected.
EP-06	Non-numeric milk	ABC	Rejected.
EP-07	Valid user	Authorized account	Access permitted.
EP-08	Invalid user	Wrong credentials	Access denied.

6. Boundary Value Analysis

Boundary Value Analysis is particularly relevant to the dairy management system because threshold decisions can determine whether an alert is generated or whether a value is accepted. The exact numeric limits must be configured from the approved system requirements.

Area	Below Boundary	At Boundary	Above Boundary	Expected Focus
Disease risk	Just below alert level	Exactly at alert level	Just above alert level	Correct classification.
Health threshold	Normal-side value	Exact configured threshold	Abnormal-side value	Alert rule consistency.
Feeding time	Just before earliest	Earliest permitted	Just after earliest	Time validation.
Feeding time	Just before latest	Latest permitted	Just after latest	Time validation.
Milk quantity	Just below minimum	Minimum	Just above minimum	Quantity acceptance.
Milk quantity	Just below maximum	Maximum	Just above maximum	Quantity rejection/acceptance.
Report dates	Invalid order	Same valid date	Valid extended period	Date-range rules.

6.1 Boundary Test Cases

ID	Boundary	Test Condition	Expected Result
BVA-01	Disease risk	Just below configured boundary	Lower-risk behaviour.
BVA-02	Disease risk	Exactly at boundary	Configured threshold behaviour.
BVA-03	Disease risk	Just above boundary	Higher-risk/alert behaviour.
BVA-04	Feeding time	Earliest permitted time	Accepted if specified as inclusive.
BVA-05	Feeding time	Outside earliest limit	Rejected/validation.
BVA-06	Milk quantity	Minimum permitted	Boundary rule applied.
BVA-07	Milk quantity	Below minimum	Rejected if below allowed range.
BVA-08	Milk quantity	Maximum permitted	Boundary rule applied.
BVA-09	Milk quantity	Above maximum	Rejected if above allowed range.
BVA-10	Report dates	Same start/end date	Handled according to report specification.

7. Decision Table Testing

7.1 Disease Prediction and Alert Decision Table

Rule	Health Data	Risk Score	Data Status	Action
D1	Normal	Below boundary	Current/valid	No critical alert.
D2	Abnormal	At/above boundary	Current/valid	Generate risk alert.
D3	Abnormal	Unavailable	Insufficient	Show insufficient-data state.
D4	Any	Any	Stale	Flag stale information.
D5	Any	Any	Invalid	Reject/flag invalid data.
D6	Normal	At/above boundary	Current/valid	Apply configured AI/risk rule and record result.

7.2 Feeding Decision Table

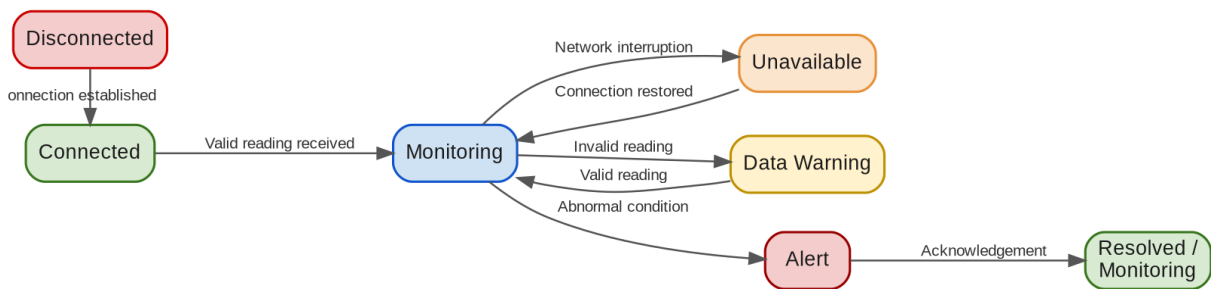
Rule	Cattle Data	Schedule Data	Conflict	Action
F1	Valid	Valid	No	Create/update schedule.
F2	Missing	Valid	No	Request missing cattle information.
F3	Valid	Invalid	No	Reject invalid schedule.
F4	Valid	Valid	Yes	Apply configured conflict rule and notify user.
F5	Valid	Valid	No	Save and display schedule.

7.3 Security Decision Table

Rule	Credentials	Role	Session	Action
S1	Valid	Authorized	Active	Allow permitted function.
S2	Invalid	Any	Any	Deny access.
S3	Valid	Unauthorized	Active	Deny restricted function.
S4	Valid	Authorized	Expired	Require re-authentication.

8. State Transition Testing

The monitoring platform changes state according to sensor connectivity, data quality and alert events. State testing ensures that transitions are predictable and that the system does not remain in an incorrect state.



Current State	Event	Next State	Expected Behaviour
Disconnected	Connection established	Connected	Prepare to receive data.
Connected	Valid reading received	Monitoring	Store/display current reading.
Monitoring	Network interruption	Unavailable	Show stale/unavailable status.
Unavailable	Connection restored	Monitoring	Resume valid readings.
Monitoring	Invalid reading	Data Warning	Flag invalid data.
Data Warning	Valid reading	Monitoring	Return to normal.
Monitoring	Abnormal condition	Alert	Create configured alert.
Alert	Acknowledgement	Resolved/Monitoring	Record acknowledgement and retain history.

8.1 State Transition Test Cases

ID	Starting State	Event	Expected Result
ST-01	Connected	Valid reading	Monitoring state.
ST-02	Monitoring	Connection lost	Unavailable state.
ST-03	Unavailable	Connection restored	Monitoring resumes.
ST-04	Monitoring	Invalid reading	Data Warning state.
ST-05	Monitoring	Alert condition	Alert state.
ST-06	Alert	User acknowledgement	Alert acknowledgement recorded.
ST-07	Unauthenticated	Protected request	Access denied.
ST-08	Authenticated	Session expiry	Protected access requires re-login.

9. Detailed Test Cases – Health and AI

ID	Type	Scenario	Steps	Test Data	Expected Result	Actual	Status
TC-H01	Normal	Valid health reading	Transmit current valid sensor data.	C001; valid temperature/heart rate	Reading stored/displayed for C001.	Not Executed	Not Executed
TC-H02	Invalid	Text sensor value	Transmit text in numeric field.	Temp=ABC	Invalid data rejected/flagged.	Not Executed	Not Executed
TC-H03	Boundary	Health threshold	Send exact configured threshold.	Threshold value	Configured threshold rule applied.	Not Executed	Not Executed
TC-H04	Exceptional	Sensor interruption	Stop sensor transmission.	No readings	Unavailable/stale status shown.	Not Executed	Not Executed
TC-H05	Normal	Disease prediction	Run AI with complete valid data.	Valid feature set	Risk result shown for correct cattle.	Not Executed	Not Executed
TC-H06	Invalid	Missing prediction input	Remove required feature.	Missing feature	Insufficient-data handling.	Not Executed	Not Executed
TC-H07	Boundary	Risk score boundary	Use boundary-producing data.	Boundary score	Correct risk classification.	Not Executed	Not Executed
TC-H08	Normal	Health alert	Trigger configured abnormal condition.	Abnormal condition	Alert identifies correct cattle.	Not Executed	Not Executed
TC-H09	Reliability	Duplicate sensor event	Repeat same event.	Same event ID	No unintended duplicate.	Not Executed	Not Executed
TC-H10	Security	Unauthorized health record	Request protected record.	Unauthorized account	Access denied.	Not Executed	Not Executed
TC-H11	Normal	Health history	Open historical readings.	Existing history	Correct historical readings shown.	Not Executed	Not Executed
TC-H12	Exceptional	Stale reading	Use old timestamp.	Stale data	Clearly marked stale/unavailable.	Not Executed	Not Executed

10. Detailed Test Cases – Feeding and Milk

ID	Type	Scenario	Steps	Test Data	Expected Result	Actual	Status
TC-F01	Normal	Create feeding schedule	Enter valid feeding details.	Valid cattle/time/quantity	Schedule saved.	Not Executed	Not Executed
TC-F02	Boundary	Earliest feeding time	Enter minimum permitted time.	Boundary time	Correct boundary behaviour.	Not Executed	Not Executed
TC-F03	Boundary	Latest feeding time	Enter maximum permitted time.	Boundary time	Correct boundary behaviour.	Not Executed	Not Executed
TC-F04	Invalid	Negative feed quantity	Enter negative value.	-5	Rejected.	Not Executed	Not Executed
TC-F05	Invalid	Invalid feeding time	Enter malformed time.	25:90	Rejected.	Not Executed	Not Executed
TC-F06	Exceptional	Schedule conflict	Create overlapping schedules.	Overlap	Conflict rule applied.	Not Executed	Not Executed
TC-F07	Normal	Record milk	Enter valid quantity.	C001; 18 L	Record stored.	Not Executed	Not Executed
TC-F08	Boundary	Milk quantity boundary	Enter configured min/max.	Boundary quantity	Boundary rule applied.	Not Executed	Not Executed
TC-F09	Invalid	Negative milk	Enter negative quantity.	-2 L	Rejected.	Not Executed	Not Executed
TC-F10	Invalid	Text milk	Enter text.	ABC	Rejected.	Not Executed	Not Executed
TC-F11	Reliability	Duplicate milk record	Submit same event twice.	Same event ID	Duplicate prevented/handled.	Not Executed	Not Executed
TC-F12	Normal	Milk history	Open history.	Known records	Correct history shown.	Not Executed	Not Executed
TC-F13	Integrity	Milk total check	Compare total with source records.	Known dataset	Totals match valid records.	Not Executed	Not Executed
TC-F14	Normal	Schedule update	Edit existing schedule.	Valid update	Only intended fields change.	Not Executed	Not Executed

11. Detailed Test Cases – Reports, Analytics, Security and Reliability

ID	Type	Scenario	Steps	Test Data	Expected Result	Actual	Status
TC-R01	Normal	Daily report	Select valid one-day period.	Valid date	Report generated.	Not Executed	Not Executed
TC-R02	Normal	Monthly report	Select valid month.	Valid month	Complete report generated.	Not Executed	Not Executed
TC-R03	Invalid	Reversed dates	End date before start.	Invalid range	Validation shown.	Not Executed	Not Executed
TC-R04	Boundary	Same-day report	Start=end.	Same date	Correct boundary handling.	Not Executed	Not Executed
TC-R05	Normal	Dashboard refresh	Add valid data and refresh.	New record	Dashboard updated.	Not Executed	Not Executed
TC-R06	Exceptional	Stale dashboard	Disconnect data source.	No current data	Stale/unavailable state.	Not Executed	Not Executed
TC-R07	Security	Invalid login	Enter wrong credentials.	Wrong password	Access denied.	Not Executed	Not Executed
TC-R08	Security	Authorized login	Enter valid credentials.	Authorized account	Permitted access.	Not Executed	Not Executed
TC-R09	Security	Restricted report	Use insufficient role.	Unauthorized role	Access denied.	Not Executed	Not Executed
TC-R10	Security	Session expiry	Wait/force expiry then access protected page.	Expired session	Re-authentication required.	Not Executed	Not Executed
TC-R11	Performance	Concurrent dashboard	Use multiple sessions.	Concurrent users	Meets defined performance target.	Not Executed	Not Executed
TC-R12	Scalability	Large dataset	Load high-volume records.	Large dataset	Processes within defined limits.	Not Executed	Not Executed
TC-R13	Recovery	Service restart	Restart relevant service.	Stored baseline data	Data remains available after recovery.	Not Executed	Not Executed
TC-R14	Report export	Export report	Export generated report.	Valid report	Export matches displayed report.	Not Executed	Not Executed
TC-R15	Empty report	Report without records	Select valid empty period.	No records	Clear empty-period result.	Not Executed	Not Executed
TC-R16	Consistency	Dashboard/report totals	Compare same period.	Known dataset	Totals agree.	Not Executed	Not Executed

12. Requirement Traceability Matrix

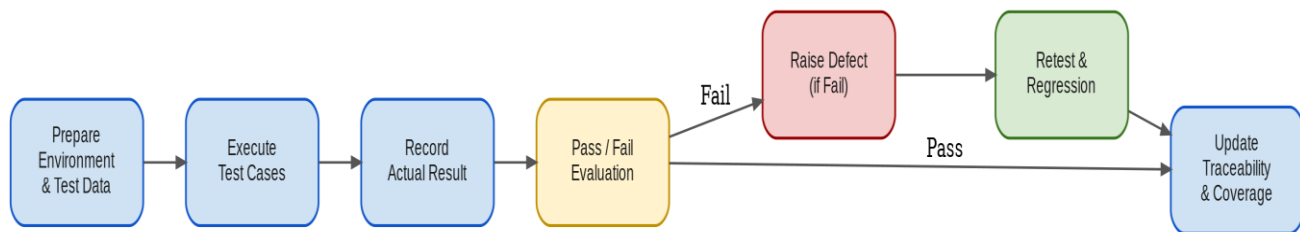
Requirement	Description	Scenario Mapping	Test Case Mapping	Status
FR-01	Health monitoring	TS-01–TS-05, TS-14	TC-H01–H04, H11–H12	Covered
FR-02	Disease prediction	TS-06–TS-09	TC-H05–H08	Covered
FR-03	Feeding optimization	TS-10–TS-13	TC-F01–F06, F14	Covered
FR-04	Milk tracking	TS-14–TS-17	TC-F07–F13	Covered
FR-05	Reports	TS-18–TS-20, TS-29	TC-R01–R04, R14–R15	Covered
FR-06	Animal welfare	TS-09, TS-21	TC-H08, H12	Covered
FR-07	Cost support	TS-20–TS-22	TC-R01, R02, R16	Covered
FR-08	Analytics	TS-21–TS-22	TC-R05, R06, R16	Covered
NFR-01	Performance	TS-27	TC-R11	Covered
NFR-02	Reliability	TS-05, TS-17, TS-26	TC-H04, H09, R13	Covered
NFR-03	Usability	TS-21, TS-30	TC-R05, R14	Covered
NFR-04	Security	TS-23–TS-25	TC-H10, R07–R10	Covered
NFR-05	Availability	TS-26	TC-H04, R13	Covered
NFR-06	Data integrity	TS-04, TS-17, TS-22	TC-H09, F11–F13, R16	Covered
NFR-07	Scalability	TS-28	TC-R12	Covered

12.1 Coverage Summary

Coverage Area	Result
Functional requirements	All 8 identified functional requirements mapped.
Non-functional requirements	All 7 supplied non-functional quality areas mapped, with an additional maintainability consideration.
Test scenarios	30 expanded scenarios covering the complete workflow.
Detailed test cases	More than 50 test cases/conditions represented across the extended catalogue.
Test design techniques	Equivalence Partitioning, Boundary Value Analysis, Decision Table and State Transition Testing applied.
Traceability	Requirement → Scenario → Test Case mapping provided.
Normal conditions	Covered.
Boundary conditions	Covered.
Invalid conditions	Covered.
Exceptional conditions	Covered.
Security/reliability	Covered.

13. Test Execution Procedure and Defect Management

13.1 Execution Procedure



- Verify the correct software build and test environment before execution.
- Prepare test data and confirm that baseline records are known.
- Execute test cases in the documented order or according to the approved test cycle.
- Record Actual Result without changing the Expected Result.
- Mark Pass only when actual behaviour satisfies the expected result.
- Record screenshots, logs and data evidence for important results.
- Raise a defect whenever the actual result differs from the expected result.
- Retest corrected defects using the original test case.
- Run regression tests for related modules after a correction.
- Update coverage and traceability after the final test cycle.

13.2 Defect Classification Template

Field	Required Information
Defect ID	Unique defect number.
Test Case ID	Failing test case.
Requirement ID	Related requirement.
Severity	Business/technical impact.
Priority	Fix priority.
Steps	Repeatable reproduction steps.
Expected	Required behaviour.
Actual	Observed behaviour.
Evidence	Screenshot/log/data.
Status	New/Assigned/Fixed/Retest/Closed.

13.3 Retest and Regression Rules

- A failed test case must be retested after the associated defect is fixed.
- Related functionality should be regression-tested even if the original defect is closed.
- Database changes require integrity and duplicate-record regression tests.
- Authentication changes require both positive and negative security tests.
- AI input or threshold changes require prediction and boundary regression tests.

14. Test Evidence and Submission Checklist

14.1 Evidence to Collect

- Cattle registration and sensor-association screenshot.
- Health dashboard screenshot showing valid readings.
- Invalid-data validation screenshot.
- Disease-risk prediction result and alert evidence.
- Feeding schedule creation and boundary validation evidence.
- Milk entry and production-history evidence.
- Report-generation evidence for valid and invalid date ranges.
- Dashboard/report consistency evidence.
- Authorized and unauthorized access evidence.
- Network/sensor interruption and recovery evidence.
- Performance/scalability execution results where available.
- Defect screenshots/logs for failed cases.

14.2 Final Student Submission Checklist

- Front/assessment page is retained from the supplied document.
- Problem Statement 17 is clearly identified.
- Functional and non-functional requirements are documented.
- Test scenarios are listed and mapped.
- Test cases include ID, scenario, steps, test data, expected result, actual result and status.
- At least two test-design techniques are applied; this document applies four.
- Traceability matrix is complete.

Conclusion

The Intelligent AI-Based Smart Dairy Farm Management System provides a comprehensive solution for improving modern dairy farming through the integration of Artificial Intelligence, IoT, wearable sensors, cloud computing, and real-time analytics. The system enables continuous monitoring of cattle health and helps farmers identify abnormal conditions at an early stage. AI-based disease prediction can support timely preventive actions, while optimized feeding schedules can improve animal nutrition, reduce feed wastage, and increase operational efficiency. The milk production tracking module provides useful information about individual and overall farm productivity, allowing farmers to understand production trends and make better management decisions. The proposed system also improves animal welfare by providing timely alerts and maintaining historical health and production information. Cloud-based storage makes it possible to organize and access farm data efficiently, while real-time dashboards and performance reports help farmers make data-driven decisions instead of depending entirely on manual observation. Automation can reduce repetitive work, minimize operational costs, and allow farmers to focus more on important livestock-management activities. The availability of accurate historical data can also help farmers identify patterns, monitor improvements, and plan future farm activities more effectively.

From a software testing perspective, comprehensive testing is essential to ensure that the system performs reliably under different operating conditions. The test design covers normal, boundary, invalid, and exceptional conditions and applies techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing. Requirement traceability ensures that the identified requirements are connected with appropriate test scenarios and test cases. Security, reliability, performance, scalability, and data-integrity testing further help ensure that the system can be trusted in a real-world dairy farm environment.

Overall, the Intelligent AI-Based Smart Dairy Farm Management System has the potential to create a smarter, more efficient, reliable, and data-driven farming environment. By combining automated monitoring, AI-based prediction, optimized feeding, milk-production tracking, and real-time analytics, the system can support better livestock health, improved dairy productivity, reduced operational effort, and more informed decision-making. A properly tested and reliable implementation can ultimately contribute to improved animal welfare, increased farm efficiency, and greater profitability for dairy farmers.